

4 (a) (i) Maximum power $P = 5.62 \text{ W}$ when $V = 4.5 \text{ V}$.

$$\text{Current } I = \frac{P}{V} = \frac{5.62}{4.5} = 1.25 \text{ A}$$

$$\text{(ii) Resistance } R = \frac{V}{I} = \frac{4.5}{1.25} = 3.6 \, \Omega \text{ (shown)}$$

$$\text{(b) } V = E - Ir$$

$$4.5 = E - 1.25r \text{ ----- (1)}$$

$$(1.60)(2.03) = E - 1.60r \text{ ----- (2)}$$

$$r = 3.6 \, \Omega$$